CLAIMS

1. A radon management system using a radon detector, the radon management system comprising:

a plurality of radon detectors (100-1 to 100-N) installed in a specific space indoors or outdoors, each radon detector (100-1, …, 100-N) configured to detect in real time alpha particles present in the specific space, output a predetermined alpha particle detection signal, count for a measurement time preset based on the output alpha particle detection signal to calculate an alpha particle concentration value, and transmit unique device identification information together with the calculated alpha particle concentration value; and

a radon management server (200) configured to collect the unique device identification information together with the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) during a preset measurement period, calculate and quantify an average value of the collected alpha particle concentration values, convert the average value into a database (DB) for each radon detector (100-1, …, 100-N), store and manage the average value, compare the alpha particle concentration value transmitted in real time from each radon detector (100-1, …, 100-N) and the average value of the previously stored alpha particle concentration values with each other, calculate an amount of change in the alpha particle concentration value, and generate radon generation event information data according to the amount of change in

the calculated alpha particle concentration value, wherein

the radon management server generates the radon generation event information data when the calculated amount of change in the alpha particle concentration value is greater than a preset reference amount of change, and wherein

the radon management server (200) converts the user action mission information corresponding to the amount of change in the calculated alpha particle concentration value into the database (DB) for each amount of change in the alpha particle concentration value so as to reduce the amount of change in the alpha particle concentration value and include information of action mission contents and action mission execution time, stores and manages the user action mission information in a separate storage device,

wherein the radon management server (200) extracts the amount of change in the alpha particle concentration value at a time of generating the radon generation event information data, reads, from the storage device, the user action mission information corresponding to the amount of change in the extracted alpha particle concentration value, and determines that the action mission contents included in the read user action mission information has been performed by a user, when the alpha particle concentration value transmitted in real time from each radon detector (100-1, …, 100-N) within the action mission execution time included in the read user action mission information is less than the average value of the previously stored alpha particle concentration values.

2. The radon management system of claim 1, wherein the radon management

server (200) transmits the generated radon generation event information data to at least one external terminal (20) through a communication network (10).

3. The radon management system of claim 1, wherein the radon management server (200) converts user action mission information corresponding to the amount of change in the calculated alpha particle concentration value into the database (DB) for each amount of change in each alpha particle concentration value, stores and manages the user action mission information in a separate storage device, extracts the amount of change in the alpha particle concentration value at a time of generating the radon generation event information data, reads, from the storage device, the user action mission information corresponding to the amount of change in the extracted alpha particle concentration value, uses the read user action mission information, and generates the radon generation event information data.

4. The radon management system of claim 3, wherein the user action mission information comprises information of action mission contents and action mission execution time so as to reduce the amount of change in the alpha particle concentration value.

5. The radon management system of claim 1, wherein, when collecting the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) during the preset measurement period, the radon management server (200) determines

the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) as noise in a case where the alpha particle concentration value is greater than a preset maximum alpha particle concentration value or less than a preset minimum alpha particle concentration value, uses only the alpha particle concentration value excluding the alpha particle concentration value determined as the noise among the collected alpha particle concentration values, and calculates the average value of the alpha particle concentration values.

6. The radon management system of claim 5, wherein, when it is determined that the action mission contents included in the read user action mission information has been performed by the user, the radon management server (200) provides a preset reward on the basis of information on the corresponding user.

7. The radon management system of claim 1, wherein the radon management server (200) collects the unique device identification information together with the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) during a preset first measurement period, analyzes the alpha particle concentration values collected during the first measurement period, extracts a first alpha particle concentration pattern, converts the extracted first alpha particle concentration pattern into the database (DB) for each radon detector (100-1, …, 100-N), and stores and manages

the first alpha particle concentration pattern.

8. The radon management system of claim 7, wherein the radon management server (200) collects the unique device identification information along with the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) during a preset second measurement period, analyzes the alpha particle concentration values collected during the second measurement period, extracts a second alpha particle concentration pattern, determines that a change has occurred in a periodic action pattern of the corresponding user in a specific place when the second alpha particle concentration pattern does not match the first alpha particle concentration pattern, generates a notification message, and then transmits the generated notification message to a preset external terminal (20) held by the corresponding user.

9. The radon management system of claim 8, wherein the radon management server (200) converts user action recommendation information corresponding to the average value of the alpha particle concentration values collected during the second measurement period into the database (DB) for each average value of the alpha particle concentration values, stores and manages the user action recommendation information in a separate storage device, calculates the average value of the alpha particle concentration values collected during the second measurement period when the second alpha particle concentration pattern does not match the first alpha particle concentration pattern, reads, from the storage device, the user action recommendation information corresponding to the average value of the calculated alpha particle concentration values,

and generates the notification message by using the read user action recommendation information.

10. The radon management system of claim 1, further comprising:

a radon reduction facility (300-1) including at least one discharging fan that suctions air entering a specific space through a vent pipe connected to the specific space, discharges the suctioned air to an external space, and reduces concentration of the alpha particles contained in the suctioned air,

wherein the radon management server (200) collects the unique device identification information together with the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) during a preset first measurement period, analyzes the alpha particle concentration values collected during the first measurement period, extracts a first alpha particle concentration pattern, converts the extracted first alpha particle concentration pattern into the database (DB) for each radon detector (100-1, …, 100-N), stores and manages the first alpha particle concentration pattern, collects the unique device identification information along with the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) during a preset second measurement period, analyzes the alpha particle concentration values collected during the second measurement period, extracts a second alpha particle concentration pattern, determines that a change has occurred in a periodic action pattern of the corresponding user in the specific place when the second alpha particle concentration pattern does not match the first alpha particle concentration pattern, and controls an operation of the radon reduction facility (300-1).

11. The radon management system of claim 10, wherein the radon management server (200) transfers a request message asking whether a change has occurred in the periodic action pattern of the corresponding user to a preset external terminal (20) held by the corresponding user when it is determined that the change has occurred in the periodic action pattern of the corresponding user in the specific place, and controls each radon reduction facility (300-1) to operate until a value where the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) is less than the alpha particle concentration value obtained before the change in the corresponding user's periodic action pattern occurs is detected when there is no response to the request message during a preset response time from the preset external terminal (20) or when receiving a response notifying that the change has occurred in the user's periodic action pattern.

12. The radon management system of claim 11, wherein the radon management server (200) determines that the radon reduction facility (300-1) is not operating properly when receiving the response notifying that no change has occurred in the user's periodic action pattern from the preset external terminal (20), generates a warning message notifying that the radon reduction facility (300-1) is in an abnormal state, and transmits the warning message to the preset external terminal (20) through the communication network.

13. The radon management system of claim 12, wherein the radon management server (200) determines that the radon reduction facility (300-1) is not operating properly when the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) after the preset radon reduction reference time is unable to become a value which is less than the alpha particle concentration value obtained before the change of the corresponding user's periodic action pattern occurs, determines that the radon reduction facility (300-1) is not operating properly, generates a warning message notifying that the radon reduction facility (300-1) is in an abnormal state, and transmits the warning message to the preset external terminal (20) through the communication network.

14. The radon management system of claim 1, wherein the preset measurement period is set as a period of measurement every 10 minutes for an hour, and

the radon management server collects the unique device identification information along with the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) during the preset measurement period, learns and analyzes the collected alpha particle concentration values, generates a radon concentration pattern over time, converts the radon concentration pattern into the database (DB) for each radon detector (100-1, …, 100-N), stores and manages the radon concentration pattern in a separate storage device, generates a push alarm message for notifying a radon concentration risk when the alpha particle concentration value transmitted in real time from each radon detector (100-1, …, 100-N) is greater than a value of the radon concentration pattern previously stored in the storage device or greater than the preset radon concentration risk reference value, and transmits the push alarm message to a

preset external terminal (20) through the communication network.

15. The radon management system of claim 14, wherein the push alarm message for notifying the radon concentration risk comprises ventilation time information corresponding to the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N).

16. The radon management system of claim 1, wherein the radon management server (200) provides a cloud computing service in response to a request from at least one client terminal, provides a cloud web service so as to download, through each client terminal, radon management-related information data converted into the database (DB) and stored in a separate storage device, encrypts and stores the radon management-related information data by using a symmetrical or asymmetric encryption method in a case of converting the radon management-related information into the database (DB) so as to be stored in the storage device, and provides a service so that a decryption key capable of decrypting the encrypted radon management-related information data is transmitted to each client terminal, and

each client terminal searches in real time the radon management-related information data stored in the storage device of the radon management server (200) by using the cloud web service of a client member login of the radon management server, displays the radon management-related information data on a display screen, decrypts the radon management-related information data encrypted and stored in the storage

device of the radon management server (200) by using the corresponding client member login information together with the decryption key transmitted from the radon management server (200), searches in real time and displays the radon management-related information data on the display screen, searches in real time the radon management-related information data stored in the storage device of the radon management server (200) through a radon management-related application downloaded from the radon management server (200), and displays the radon management-related information data on the display screen.

17. The radon management system of claim 1, wherein, based on the unique device identification information together with the alpha particle concentration value transmitted from each radon detector (100-1, …, 100-N) or the average value of the calculated alpha particle concentration values, the radon management server (200) provides a service for the radon management-related information data to be displayed on a display screen of an administrator or displayed on the user's display screen provided in an external terminal (20), so that the administrator is able to monitor the radon management-related information data for each radon detector (100-1, …, 100-N) on a daily, day of the week, weekly, monthly, quarterly, and yearly basis, and on the basis of a real-time GIS (Geographic Information System).